

1 Causes of variation in vaccine efficacy in a wild rodent
2 population

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13 **Individuals vary broadly in their response to vaccination and subsequent chal-**
14 **lenge infection, with poor vaccine responders causing persistence of both**
15 **infection and transmission in populations.**

16 **Keywords:** Helminth; vaccine; immunoparasitology; machine learning; pathway ana-
17 lysis; ageing.

INTRODUCTION

18 Individuals vary widely in their responses to vaccination and little is known about the
19 causes underlying low vaccine efficacy, nor how effective a vaccine will be “in the real
20 world” against the pathogen for which it has been developed.¹

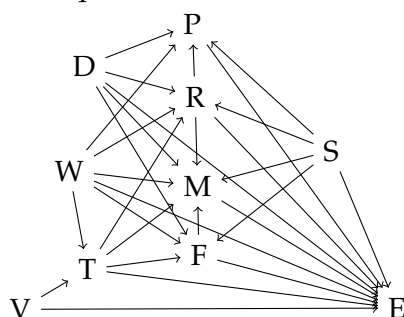
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RESULTS

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METHODS

48 Notebooks produced using Pluto.jl v0.17.7²

ACKNOWLEDGEMENTS

COMPETING INTERESTS

49 The authors declare that there are no competing interests.

AUTHOR CONTRIBUTIONS

REFERENCES

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 53 [zenodo.5889169](https://doi.org/10.5281/zenodo.5889169).

FIGURE LEGENDS

SUPPLEMENTARY INFORMATION

54 **Supplementary Data 1**

55 **Supplementary Data 2**

56 **Supplementary Data 3**

57 **Supplementary Figures**